



1. What are the main features of Java?
2. Difference between JDK, JRE, and JVM.
3. Explain platform independence in Java.
4. What is bytecode in Java?
5. Difference between == and .equals() in Java.
6. Explain wrapper classes in Java.
7. What is autoboxing and unboxing?
8. Difference between final, finally, and finalize().
9. What are access modifiers in Java?
10. Explain the difference between static and non-static methods.
11. What are the four principles of OOP?
12. Difference between abstraction and encapsulation.
13. Difference between interface and abstract class.
14. Can we achieve multiple inheritance in Java? How?
15. What is method overloading vs method overriding?
16. What is constructor overloading?
17. Can a constructor be private?
18. What is super keyword in Java?
19. Difference between this and super.
20. Explain polymorphism with an example.
21. What is exception handling in Java?
22. Difference between checked and unchecked exceptions.
23. Explain try-catch-finally with an example.
24. Difference between throw and throws.
25. How to create a custom exception?
26. What is multithreading in Java?
27. Difference between process and thread.





28. What is synchronization in Java?
29. Difference between sleep() and wait().
30. What is a daemon thread?
31. Write a Java program to reverse a string.
32. Write a Java program to check if a number is prime.
33. Write a Java program to find the factorial of a number using recursion.
34. Write a Java program to check if a string is a palindrome.
35. Write a Java program to find duplicate elements in an array.
36. Write a Java program to implement a simple ATM system.
37. Write a Java program to sort an array without using built-in methods.
38. Write a Java program to find the largest and smallest numbers in an array.
39. Write a Java program to count the frequency of characters in a string.
40. Write a Java program to demonstrate multithreading (two threads printing numbers).
41. What is the difference between shallow copy and deep copy in Java?
42. Define class and objects?
43. Write a Java program to remove duplicates from an array.
44. What is difference between int and Integer?
45. Can we declare a constructor as final?
46. Difference between interface and abstract class.
47. Difference between Runnable and Thread class.
48. Difference between String, StringBuilder, and StringBuffer.
49. Can you overload the main method in Java?
50. What is type casting in Java?

